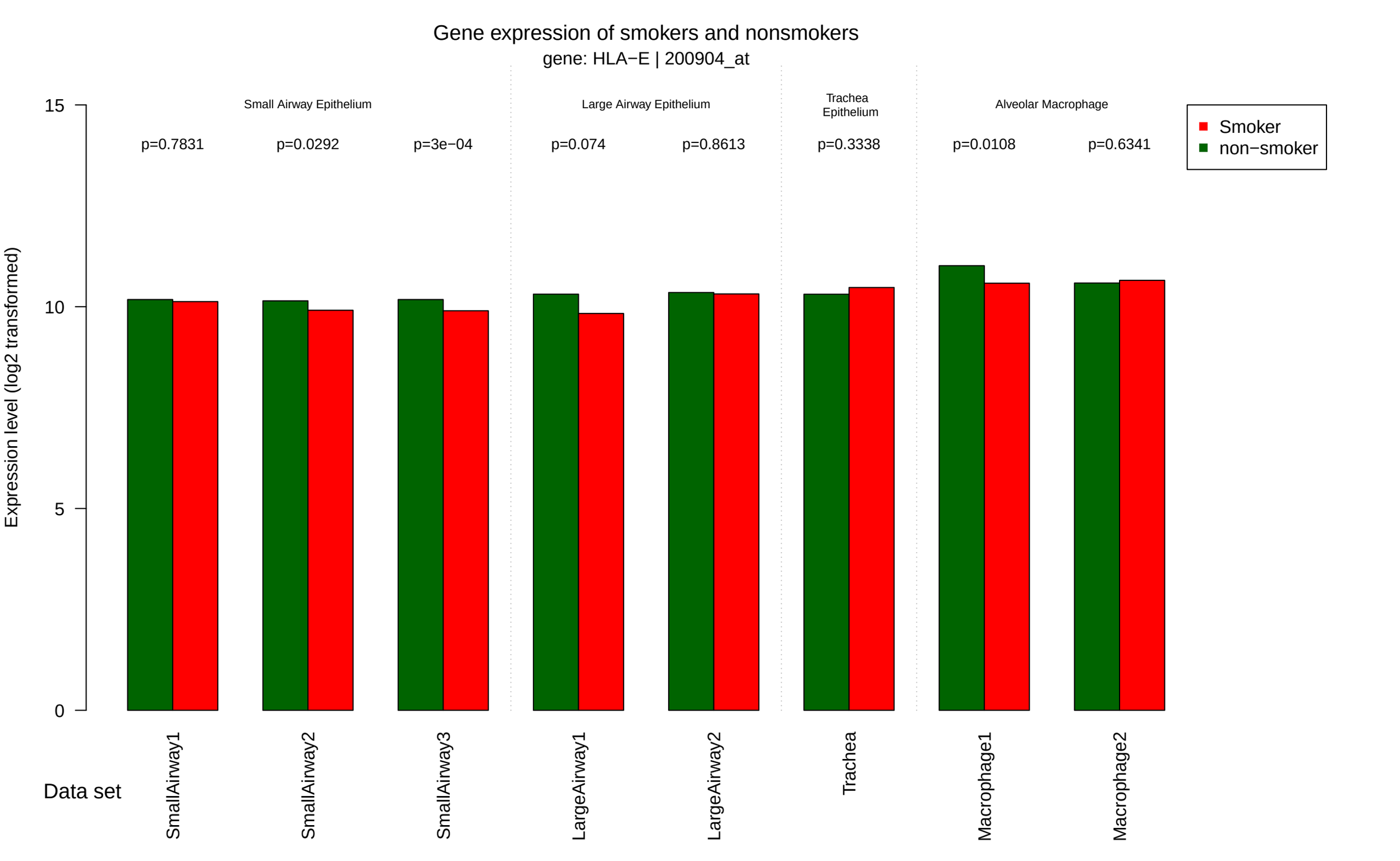
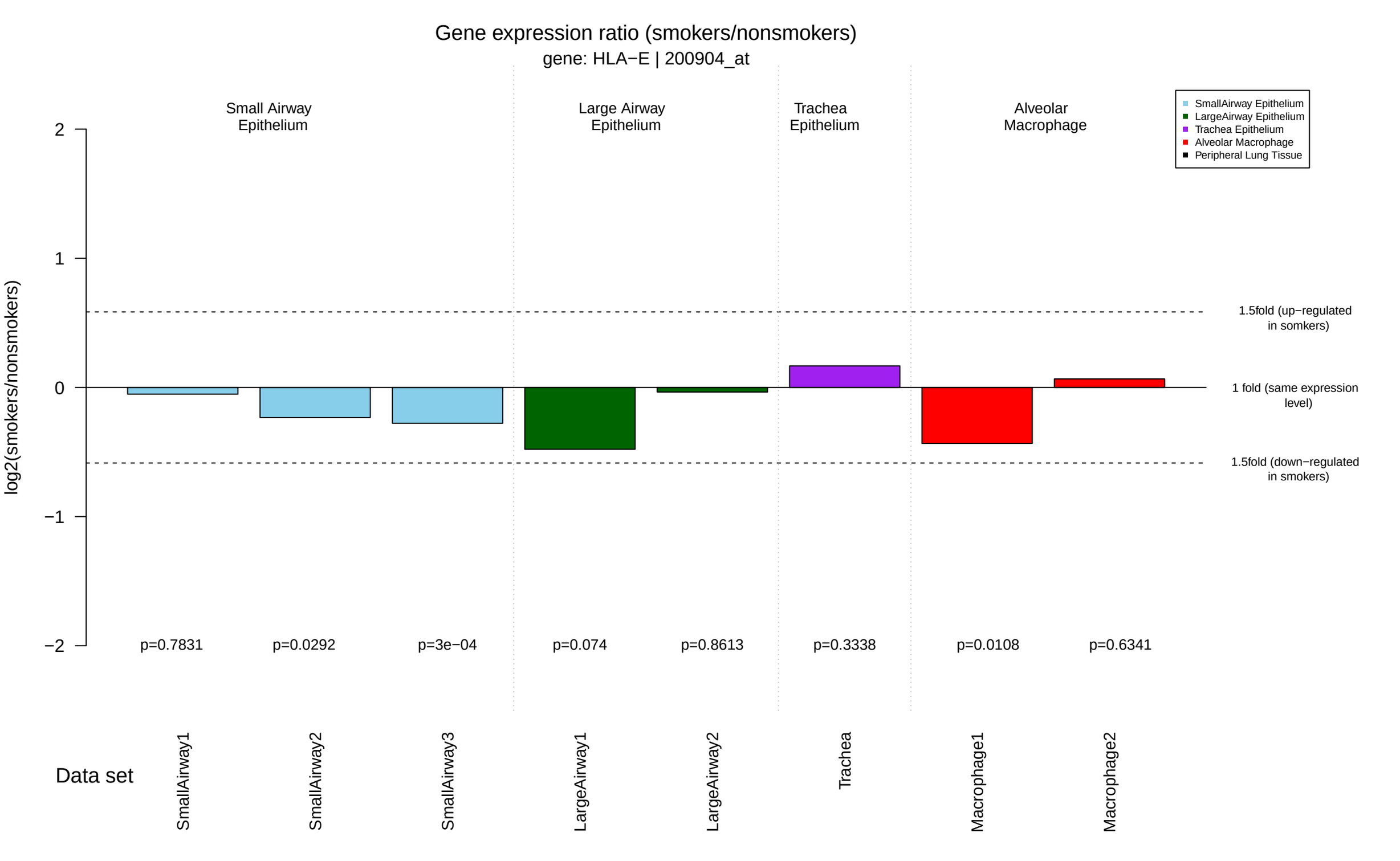
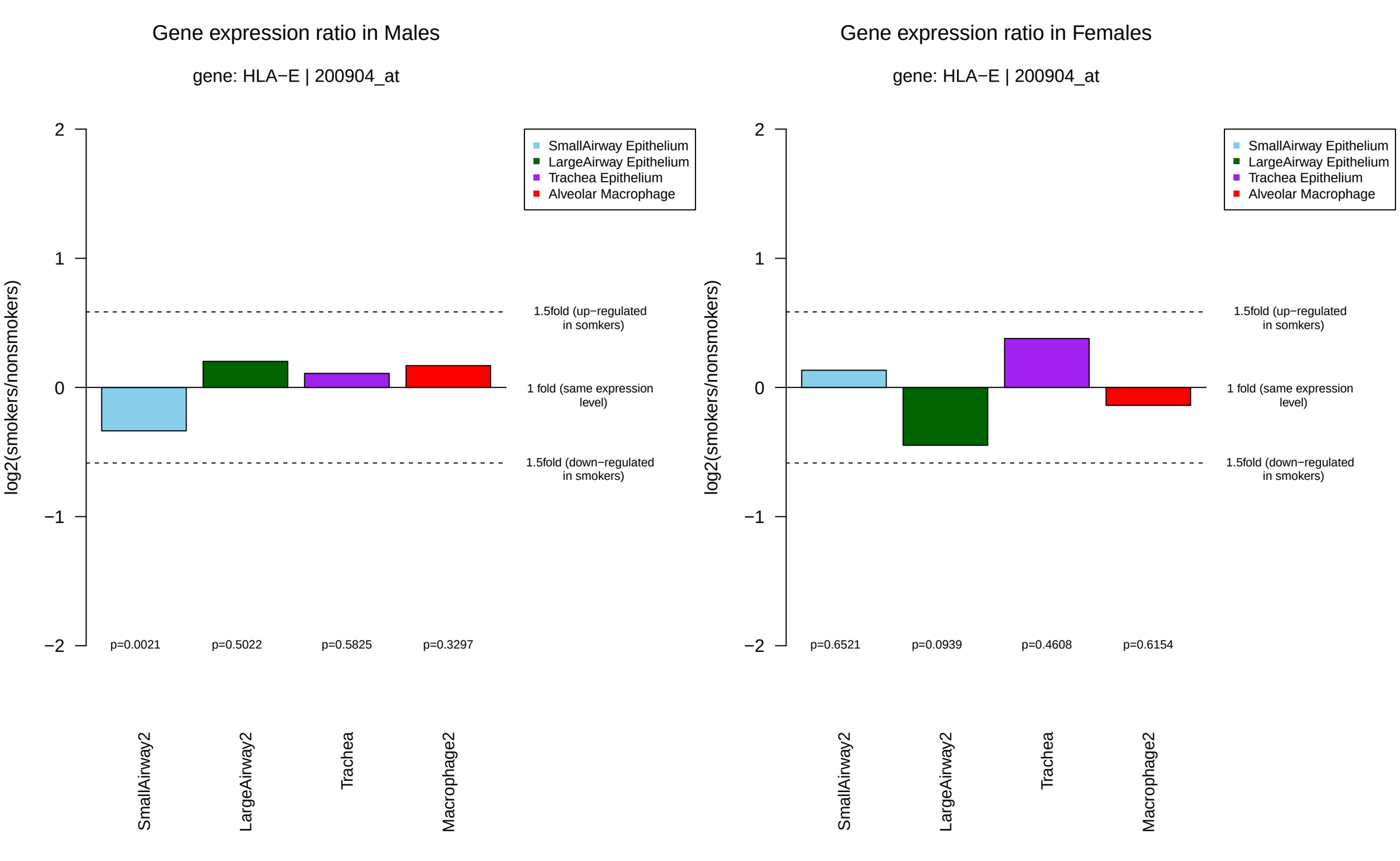
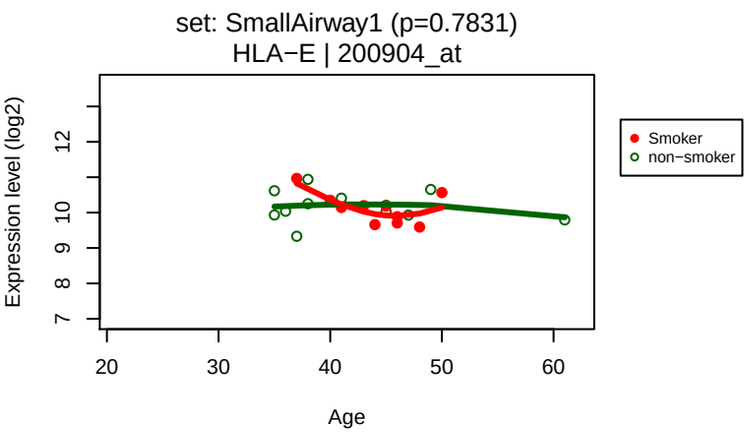
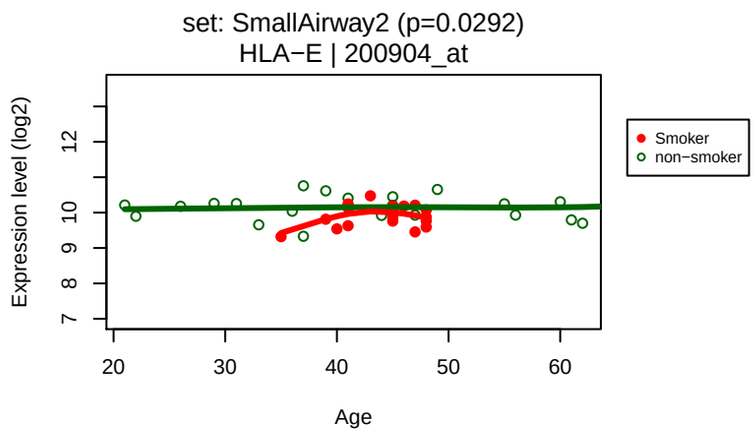
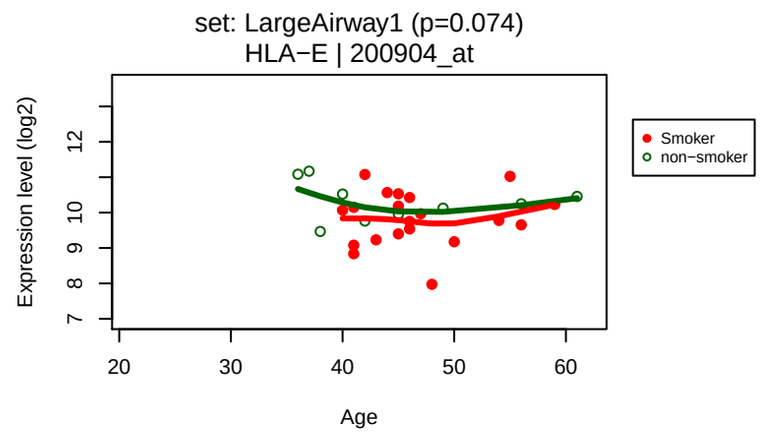

Gene expression of smokers and nonsmokers
gene: HLA−E | 200904_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7831
p=0.0292
p=3e−04
p=0.074
p=0.8613
p=0.3338
p=0.0108
p=0.6341
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

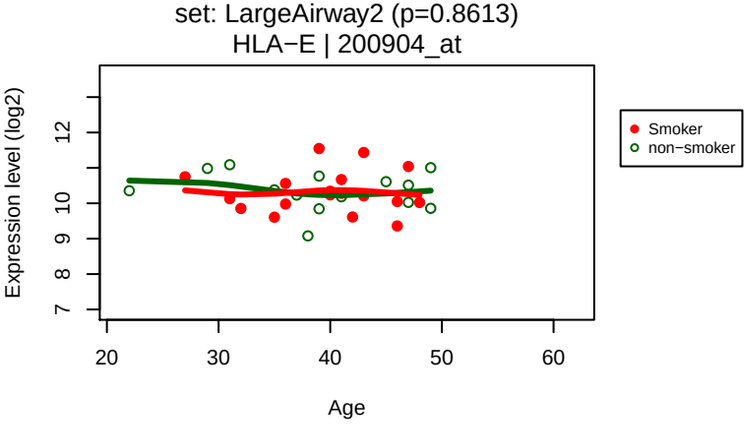

Gene expression ratio (smokers/nonsmokers)
gene: HLA−E | 200904_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7831
p=0.0292
p=3e−04
p=0.074
p=0.8613
p=0.3338
p=0.0108
p=0.6341
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

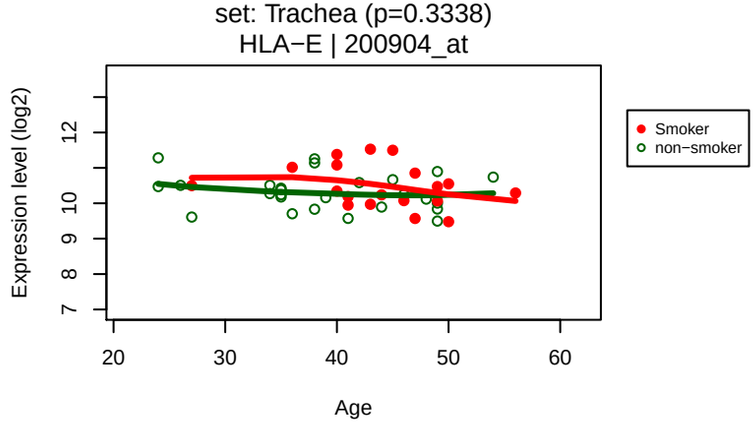

Gene expression ratio in Males
gene: HLA−E | 200904_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0021
p=0.5022
p=0.5825
p=0.3297
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−E | 200904_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6521
p=0.0939
p=0.4608
p=0.6154
SmallAirway2
LargeAirway2
Trachea
Macrophage2

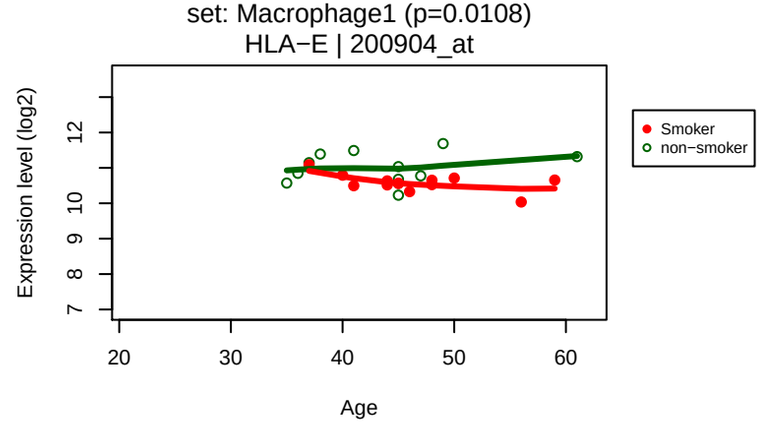

set: SmallAirway1 (p=0.7831)
HLA−E | 200904_at
Expression level (log2)
Age
Smoker
non−smoker

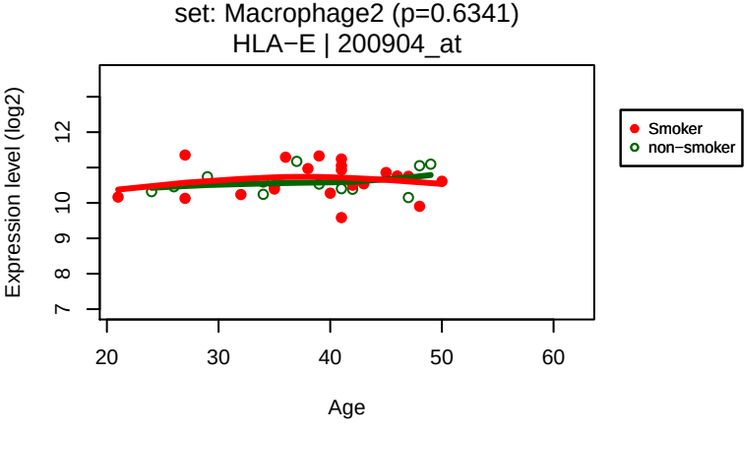

set: SmallAirway2 (p=0.0292)
HLA−E | 200904_at
Expression level (log2)
Age
Smoker
non−smoker

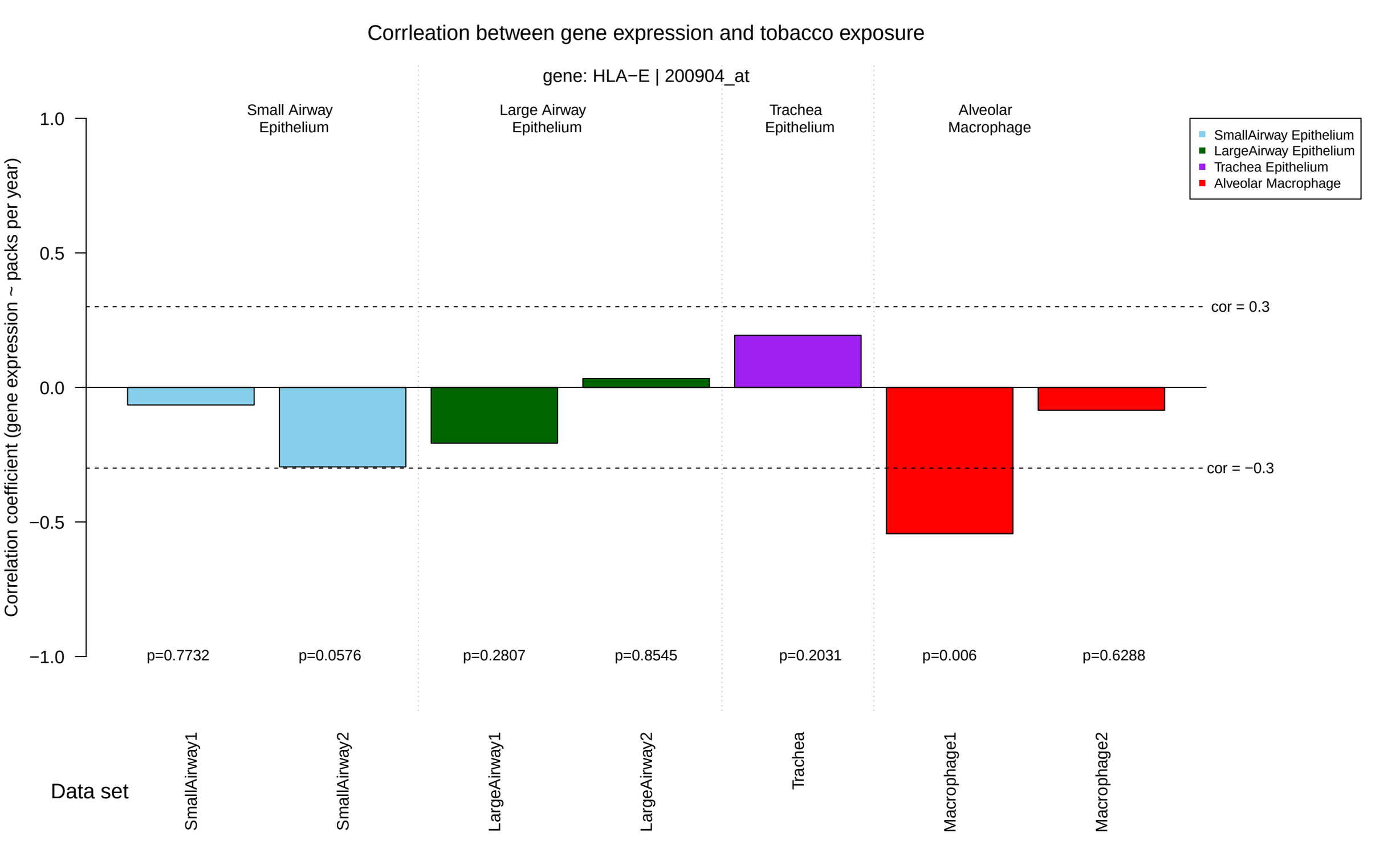

set: LargeAirway1 (p=0.074)
HLA−E | 200904_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8613)
HLA−E | 200904_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3338)
HLA−E | 200904_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0108)
HLA−E | 200904_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6341)
HLA−E | 200904_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−E | 200904_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7732
p=0.0576
p=0.2807
p=0.8545
p=0.2031
p=0.006
p=0.6288
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2